QUANT BEYOND IRR

KRITI-Inter Hostel Tech Competition

Final Submission Report

Hostel Id 2633

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1 PROBLEM STATEMENT OVERVIEW

1.1 Objective

As part of this problem statement with LevUp, a specialist in quantitative wealth management, we were tasked with designing and implementing a fully automated, long-only equity strategy for the Indian stock market using systematic, algorithm-driven methodologies. The objective was to develop a rule-based strategy grounded in quantitative models and rigorously backtest it on historical data from 2010 to 2020, ensuring robustness, consistency across market regimes, and strict elimination of look-ahead bias or any form of data leakage.

1.2 Dataset Provided

We are provided with daily OHLCV data for Indian stocks.

- **Training Period:** 2010–2020.
- **Columns:** Date, Symbol, Open, High, Low, Close, Value, Volume, Sector, Mcap, LMS, and `in_nse500` flag.
- **Benchmark Data:** A separate `indexes.xlsx` containing daily OHLC data for the Nifty 500 index, used for calculating relative performance metrics.

1.3 Portfolio & Execution Constraints

The strategy must strictly adhere to the following capital and execution rules:

Parameter	Constraint
Initial Capital	₹50,00,000 (50 Lakhs)
Portfolio Type	Long-only (No short selling, No leverage)
Positions	Min: 1 stock | Max: 100 stocks
Allocation	No fractional shares; Cash up to 100% allowed
Rebalancing	Frequency $\geq$ 1 day
Execution Timing	**T+1** (Trade executed next day after signal)
Execution Price	Average of $(Open + High + Low + Close)$ on T+1

1.4 Transaction Costs

Costs are applied automatically on every buy and sell trade.

- **Breakdown:** STT (0.10%) + GST on STT (0.018%) + Brokerage (0.10%) + Slippage (0.05%).
- **Total Cost:** $\approx$ **0.268% per side**.

2 QUANTITATIVE MARKET OVERVIEW: NIFTY 500 (2010 – 2020)

From 2010 to 2020, the Nifty 500 went through clear cycles of growth and decline, along with a few sharp market corrections.

2.1 Returns Distribution and Performance Phases

Between Jan 2010 and Jan 2020, the Nifty 500 delivered a 10-year CAGR of approximately 8.5%. Over this period, performance can be broadly grouped into three phases:

- **Contraction Phase:** 2011 recorded the sharpest annual fall (-27.2%), with smaller negative years in 2015 (-0.72%) and 2018 (-3.38%).
- **Expansion Phase:** Strong rallies were seen in 2014 (+37.8%) and 2017 (+35.9%), marked by steady upward movement and relatively stable market conditions.
- **Recovery Phase:** After the 2011 fall, the market rebounded strongly in 2012 (+31.8%), showing clear recovery behavior.

2.2 Volatility and Risk Metrics

Over the long term, annualized volatility remained fairly stable between 17.3% and 17.8%, although short-term movements varied across years:

- **Realized Volatility:** Around 12% in calmer years such as 2017, and above 22% during stressed periods like 2011 and 2018.
- **Risk-Adjusted Returns:** The Sharpe Ratio stayed moderate, partly due to relatively high risk-free interest rates between 2010 and 2014.

2.3 Drawdown and Downside Risk

Maximum drawdown analysis highlights two major stress periods during this decade:

- **2011 Correction:** A prolonged decline of nearly -30% that took more than 12 months to fully recover.
- **2018 NBFC Crisis:** A sharp fall in stock price triggered by the IL&FS default, which heavily impacted mid-cap and small-cap stocks within the index.

2.4 Liquidity and Execution Profile

The Average Daily Turnover (ADT) of Nifty 500 stocks increased from 72.5% (2010) to 93.0% (2019) of the total NSE universe. This improvement in liquidity has important implications for quantitative strategies:

- **Lower Impact Costs:** Higher liquidity reduces trading costs and allows larger capital deployment, especially in mid-cap stocks.
- **Volume-Price Relationship:** Trading volumes became more closely linked with price movements, particularly as mid-cap weights stabilized around 13–14%.

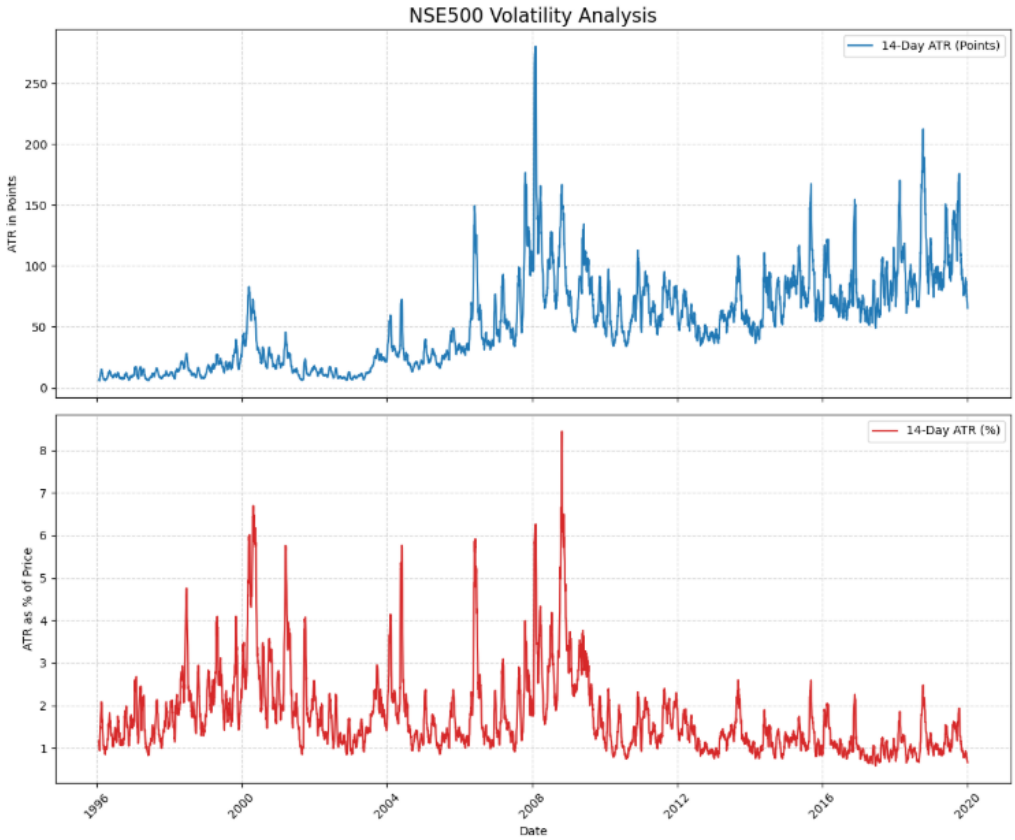

Figure 1: Volatility Analysis of the NSE500 Index (2010–2020), displayed in terms of ATR points and ATR

2.5 Conclusion

Overall, this analysis shows that the Nifty 500 experienced clear market cycles, periodic sharp drawdowns, and changing volatility conditions during 2010–2020. For strategy design, this means we must account for regime shifts, control drawdowns during stress periods, and ensure that the strategy remains robust across both high- and low-volatility environments. Liquidity considerations are also important to maintain realistic execution and scalable portfolio construction.

3 STRATEGY DESCRIPTION

In our strategy, we have divided the total initial funds our portfolio has allocated 70% to Strategy A (split 35/35 internally) and 30% to Strategy B. This trading model divides the total initial funds into a balanced 35/35/30 split for the strategies.

First, we put 70% of the funds into Strategy A. This strategy is then split into two parts: half of it acts aggressively to grow the money, and the other half acts defensively to protect it.

The remaining 30% goes to Strategy B. Its job is to find the market's current winners and hold them as long as their trend persists.

Because these two strategies use completely different formulas, look at different timeframes, and trade on different schedules, they rarely overlap or make the same mistakes at the same time. This setup keeps the money working as efficiently as possible while lowering the overall risk of the portfolio.

3.1 Strategy A: The Smooth Trend Engine (70% Allocation)

Strategy A is the steady base of our portfolio. Every five days, it looks for stocks that are growing reliably. It avoids stocks that are jumping up and down wildly or going up too fast.

3.1.1 Feature Engineering & Noise Reduction

Daily stock prices are noisy and can jump up and down randomly. To fix this, we run the prices through a tool called a Kalman Filter. This smooths out the data so we can clearly see where the stock is really going. We then use this smoothed-out price to score and rank every stock based on four main features:

1. **Kalman Momentum (M):** It tells us how much the smoothed price has grown over time and shows us the true direction the stock is heading, ignoring the noise.

2. **RSI Quality Score:** We look for stocks growing at a safe & steady speed. We stay away from stocks which are crashing down or jumping up too fast. We use a 'perfect' score of 55 to find these steady stocks. If a stock moves too wildly or too slowly, our formula gives it a lower rank to keep our money safe.
   Formula:

$$\boxed{Q_{RSI} = \max\left(0, 1 - \frac{|\text{RSI} - 55|}{55}\right)}$$

2. **Trend Quality Score:** We measure the gap between today's actual stock price and our smooth trend line. If a stock's price moves too far away from this steady line, it is considered overextended and risky because it went up too far. To protect the portfolio, our formula automatically reduces the stock's score:

$$\boxed{Q_{Trend} = \frac{1}{1 + |d_{KF}| \times 10}}$$

3. **Consistency:** Here we look at the 60-day rolling Standard Deviation of daily returns. We take its inverse so that stocks with a high Standard Dev. automatically gets a lower rank. This helps us avoid risky, unstable stocks.

Feature	Weight	Objective
Kalman Momentum	50%	Primary method to follow steady,medium-term trends.
Quality (RSI)	20%	Avoid stocks that spike too fast or drop too hard.
Trend Quality	15%	Protect against mean-reversion crashes.
Consistency	15%	Reward low-volatility, steady compounders.

Table 1: Strategy A's Scoring System Breakdown

3.1.2 Inner Split: Aggressive Defensive Sides

Once every stock gets its final score, Strategy A selects the best ones among all and splits its 70% capital equally into two different trading sections to manage the group of winning stocks:

- **Aggressive section (35% of Total Portfolio):** Capital is distributed equally across all selected stocks. This allows us to fully capture the market's strongest trends without relying too heavily on just one or two big stocks.

- **Defensive section (35% of Total Portfolio):** Capital is distributed via **inverse-volatility weighting** ($w_i \propto 1/\sigma_i$). This automatically puts less money into stocks that jump around a lot, and more money into reliable, steady growers. As a result, it naturally calms down the overall risk of our portfolio.

3.2 Strategy B: The Fast-Moving Strategy (30% Allocation)

Every week, we update Strategy B to focus on fast growth. It looks at stock returns over different time periods to find the strongest market leaders. We then put more money into the best picks to try and get the highest possible profits.

3.2.1 Tracking Trends Over Different Times

To avoid the common mistake of only looking at one time period, Strategy B focuses on long, medium and short term strength. This makes sure the system only buys stocks that are already showing a strong, proven trend.

Indicator	Weight	Rationale
12-Month Return	40%	Long-term base; confirms long-term institutional accumulation.
6-Month Return	30%	Confirms mid-term trend continuation.
1-Month Return	10%	Provides short-term entry timing.
Trend Strength	10%	Price gap over the 200-day EMA.
Volatility Percentile	10%	Small penalty for stocks that are too volatile.

Table 2: How Strategy B Scores Stocks

3.2.2 Execution Hysteresis & Dynamic Sizing

To save on fees and taxes, Strategy B avoids trading too much. We only buy a stock if it enters the Top 45 list. We don't sell it until it drops past the 55th rank or starts crashing. This method stops us from selling too early.

We don't put the same amount of money into every stock. Instead, we give more money to the best-ranked and steadiest stocks. To keep things safe, we never let any single stock take up more than 5% of our total money.

3.3 The Final Plan: How the Capital is Allocated

We faced one big challenge when combining these strategies: lazy money. Since Strategy B needs a full year of past data before it can start making decisions, setting a strict 70/30 split on Day 1 means 30% of our money would just sit there doing nothing for an entire year. To make sure every rupee is working hard from the very start, we use a flexible funding approach:

- **Phase 1 (100% Efficiency):** On deployment, 100% of the portfolio's initial capital is assigned to Strategy A, allowing the algorithm to immediately compound wealth at full capacity.
- **Phase 2 (The Transition Event):** As soon as Strategy B has enough past data and is ready to make its first trade, we check the new, total value of our account. We take exactly 30% of that total, subtract the normal trading fees for selling those shares, and hand that cash over to Strategy B to start working.
- **Phase 3 (The Final Setup):** From then on, the two strategies just run side-by-side in our perfectly balanced 70/30 split.

4 RISK MANAGEMENT FRAMEWORK

Our primary objective is capital protection. Long-term performance is only possible if we first avoid large losses. To ensure stability over a 10-year period, we use an automated risk management system that operates at three levels:

- Individual stocks - The overall portfolio - Trade execution

4.1 Stock-Level Controls

We control risk at the individual stock level using clear, rule-based safeguards:

- **Volatility-Adjusted Trailing Stops:** Stop-loss levels are adjusted based on each stock's volatility. Less volatile stocks are given tighter stop levels, while more volatile stocks are allowed slightly wider movement. The stop level moves upward with the stock price but never moves down.
- **Inverse-Volatility Position Sizing:** Capital allocation is inversely proportional to volatility ($w_i \propto 1/\sigma_i$). More volatile stocks receive smaller allocations, while stable stocks receive larger weights.
- **Position Limits:** No stock in Strategy B is allowed to exceed 5% of total NAV. If a position grows beyond this level, it is automatically trimmed and excess capital is moved to cash.

4.2 Portfolio-Level Controls

While stock-level rules manage individual risks, portfolio-level rules protect against broad market downturns. The system adjusts exposure depending on overall market conditions.

- **Moving Average Filter (Strategy A):** The Nifty 500 is compared to its 200-day moving average. If the index falls below this level, the system reduces defensive allocation from 50% to 25% and shifts the remaining capital to cash. This reduces exposure during weak market phases.
- **Market Breadth Filter (Strategy B):** The system monitors how many stocks are trading below their 200-day EMA.
  - **Bear Condition:** If more than 50% of stocks trade below their 200-day EMA, all target weights are scaled down by 0.90, and maximum portfolio capacity reduces from 50 stocks to 15.
  - **Crash Condition:** If more than 70% of stocks trade below their 200-day EMA, maximum portfolio capacity is reduced to 5 stocks.

4.3 Execution Controls

To ensure realistic results, the strategy accounts for real market constraints and avoids unrealistic assumptions.

- **Circuit Breaker Handling:** If a stock hits a lower circuit (no buyers available), the system does not assume an exit is possible. Instead, the position remains open until trading resumes, reflecting realistic drawdowns.
- **T+1 Execution and Transaction Costs:** Signals are generated using Day T closing data and executed on Day $T+1$ at the OHLC-4 average price. A transaction cost of 0.268% is deducted on every trade, including partial rebalancing.

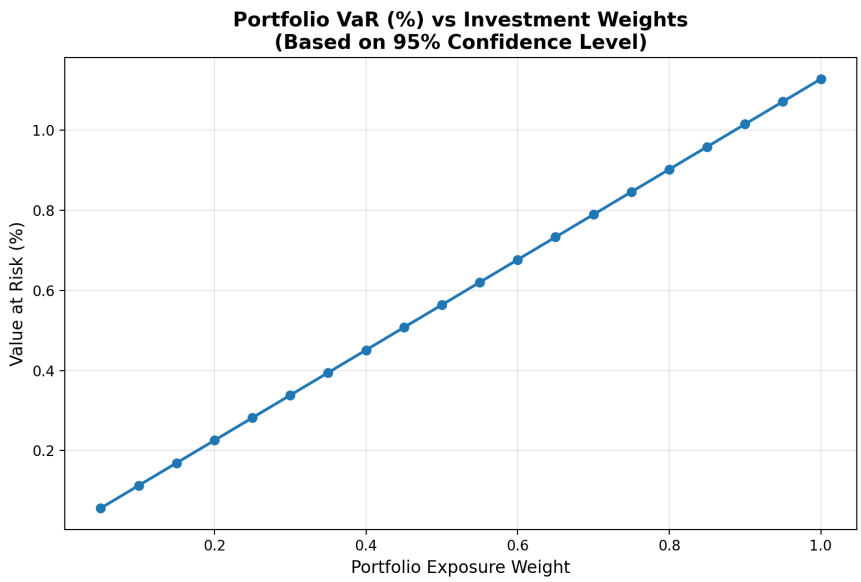

Figure 2: Daily 95% VaR compared with portfolio exposure.

4.4 Delisted Stocks Handling

Backtesting frameworks often suffer from survivorship bias by silently removing stocks that get delisted, merged, or permanently suspended. Such behavior artificially inflates performance metrics by avoiding permanent capital loss.

To eliminate this distortion, our engine explicitly detects terminal delistings.

- If a stock held in the portfolio is missing on execution day $T + 1$, the stock is forcibly liquidated using the **last available trading day's OHLC average price**, defined as:

$$\text{OHLC} = \frac{Open + High + Low + Close}{4}$$

- Transaction costs of 0.268% per side are deducted exactly as in normal sell executions.

This mechanism ensures that permanent capital impairment is properly absorbed by the portfolio and prevents artificial performance inflation caused by survivorship bias.

Importantly, The liquidation rule activates only when the stock don't appear on the execution day $T + 1$

5 PERFORMANCE METRICS

The performance of the strategy is evaluated using standard return and risk measures to assess both profitability and downside control.

- **CAGR:** The strategy achieved a Compound Annual Growth Rate of 22.37%, indicating strong long-term compounding.
- **Volatility:** Annualized volatility was 11.41%, showing relatively stable return variation for an equity strategy.
- **Sharpe Ratio:** The Sharpe Ratio of 1.86 reflects strong risk-adjusted returns.
- **Maximum Drawdown:** The largest peak-to-trough decline was -19.41%, indicating controlled downside during market stress.
- **Information Ratio:** An Information Ratio of 1.09 suggests consistent outperformance relative to the benchmark.
- **Up Capture Ratio:** The strategy captured 64.70% of market upside.
- **Down Capture Ratio:** The strategy captured 47.22% of market downside, indicating partial downside protection.

Rolling Window	Average Outperformance (%)	Worst Underperformance (%)
1 Year	16.24%	-7.15%
3 Years	76.22%	14.06%
5 Years	198.89%	60.84%

Table 3: Rolling outperformance relative to Nifty 500

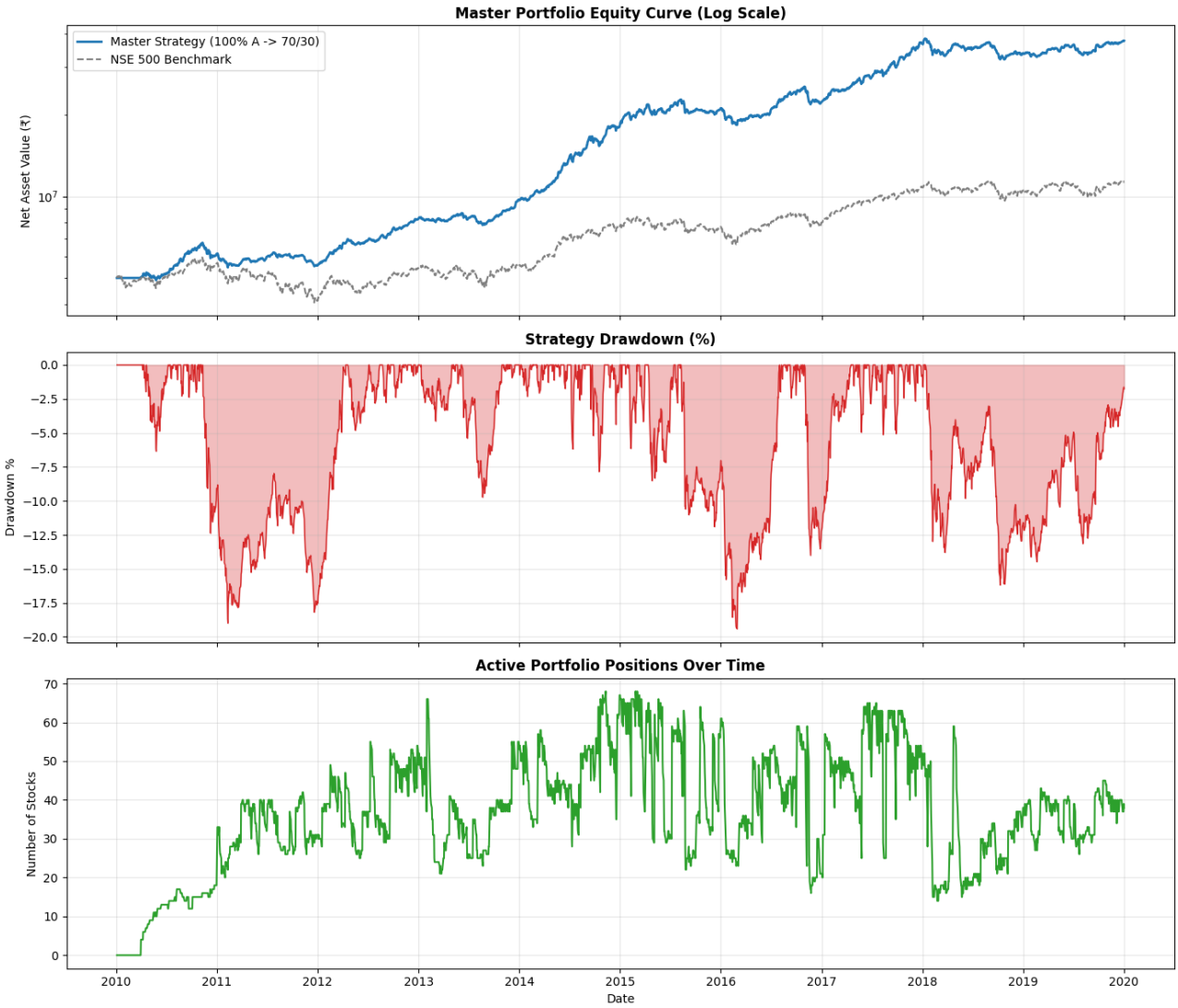

Figure 3: Equity curve (log scale), drawdown profile, and number of active positions over the evaluation period.

Figure 4: Rolling 1-year Sharpe ratio.

6 STATISTICAL VALIDATION AND ROBUSTNESS

6.1 Hypothesis Testing Results

Our trading strategy has been mathematically validated for effectiveness, ensuring that the generated returns are statistically significant and not the result of random market drift:

- **T-test Statistic:** 2.50 with a P-value of 0.01252, confirming statistical significance at the 0.05 level.
- **Mean Return:** 0.00049 with a 95% Confidence Interval of [0.00009, 0.00088], indicating reliable positive returns.
- **Win Rate:** 55.66%, with a Profit Factor of 1.19, demonstrating strong edge and risk-reward asymmetry.
- **Profitability:** Total Profit of 4.60 significantly exceeds Total Loss of 3.86, underscoring the strategy's overall structural profitability.

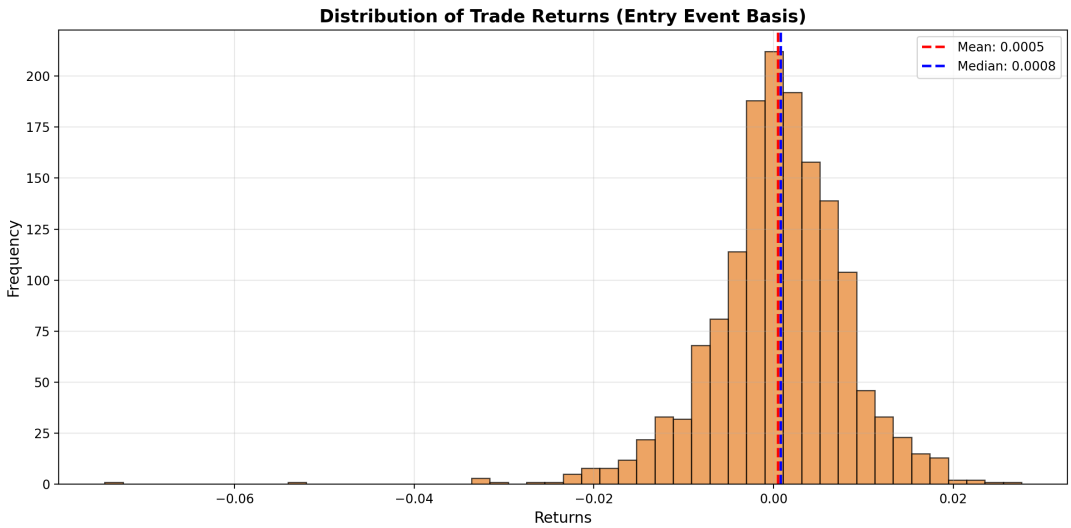

Figure 5: Distribution of the user hypothesis showing the expected variance and density across the tested parameters.

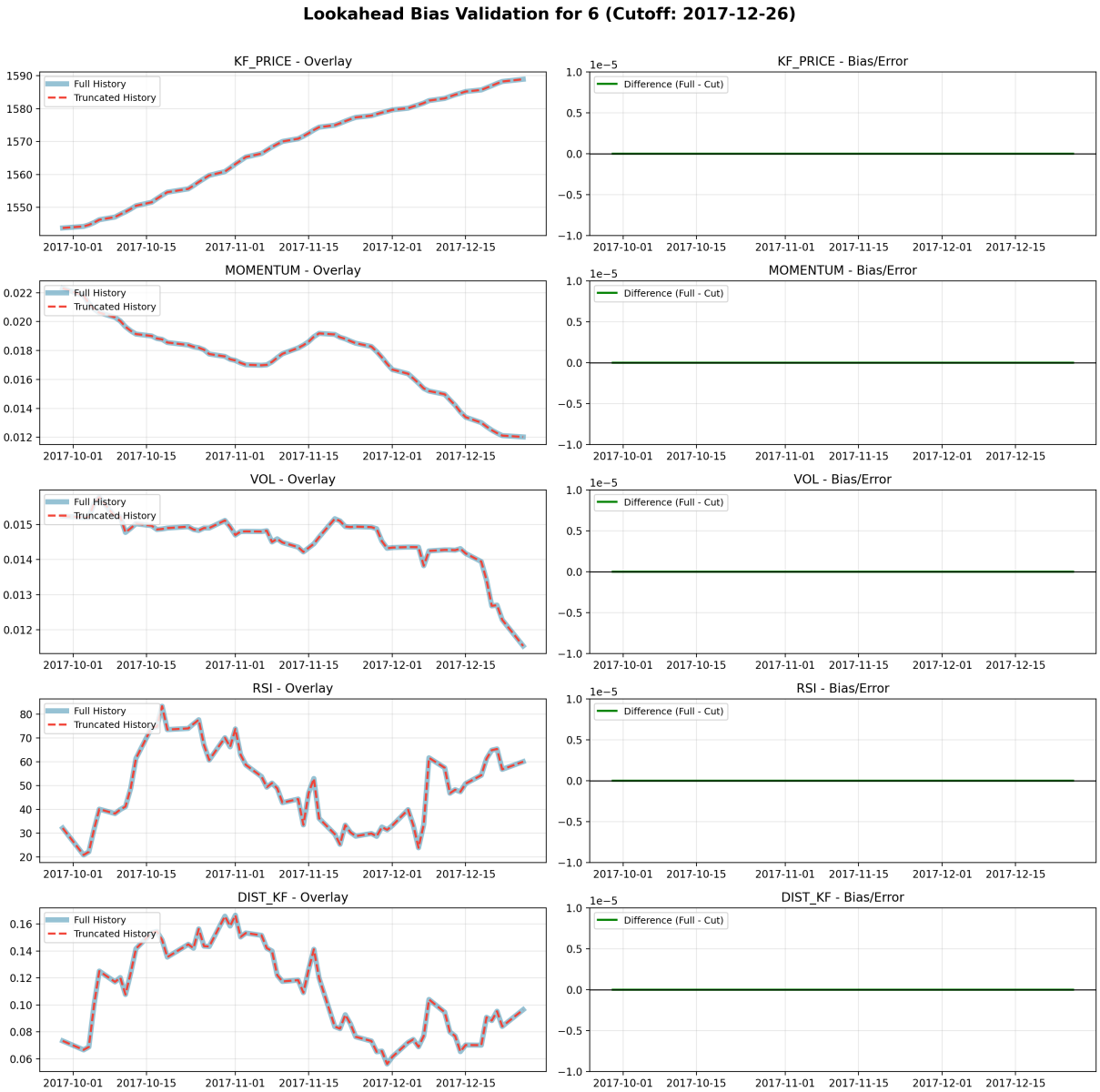

Figure 6: Lookahead Bias Validation

6.2 Monte Carlo Simulation (1,000 Paths)

To check whether the strategy's strong performance is due to a genuine edge rather than luck, we conducted a 1,000-path Monte Carlo simulation using Geometric Brownian Motion (GBM).

The results show that the strategy is highly stable. The actual final portfolio value is very close to the median outcome from the simulations, indicating that the performance is consistent with its statistical expectation. Even in the worst-case scenario (5% VaR), the portfolio still grows the initial ₹5,000,000 capital by more than 4 times. This suggests that the probability of severe downside under these assumptions is low.

- **Original Final Equity:** ₹37,581,095
- **Median Expectation:** ₹37,700,452
- **Worst Case (5% VaR):** ₹20,263,645
- **Best Case (95% Upside):** ₹69,519,989

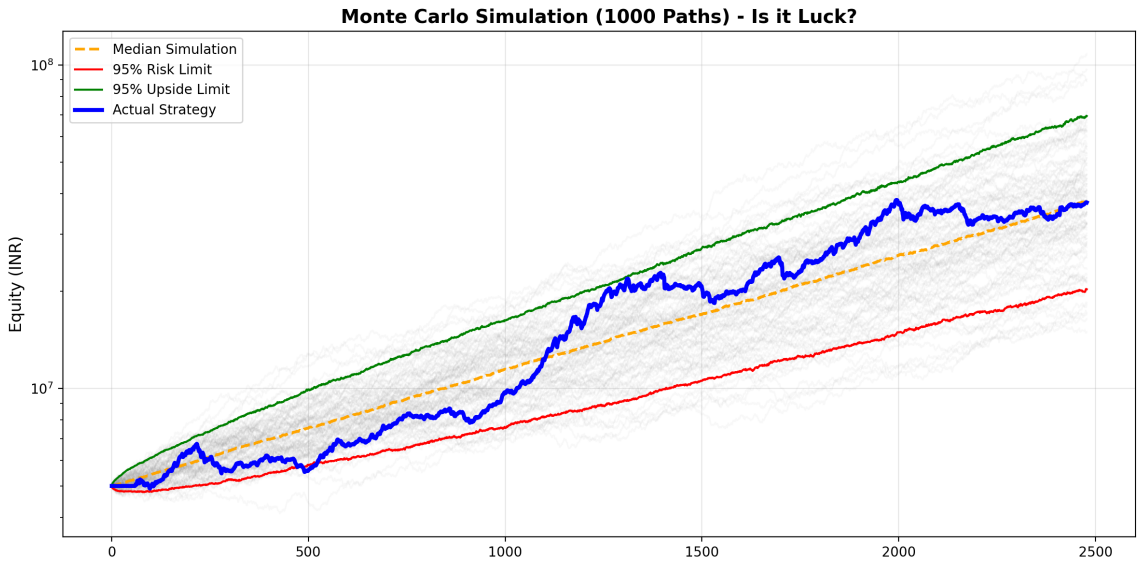

Figure 7: 1,000-run Monte Carlo simulation of the strategy's equity curve.